Milestones M5–M6: `ResidualBound.jl`

Certified $\|\mathcal{R}\|_\infty$ Bound for the Babenko Equation

`wave-cap` — Computer-Assisted Proof for the Babenko Equation

BabenkoRigorous.jl

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Abstract

We describe the implementation and validation of `ResidualBound.jl`, covering Milestones M5 (pure gravity, $\beta = 0$) and M6 (gravity-capillary, $\beta \neq 0$). The module exports `residual_bound`, a single function that returns a certified Arb upper bound on

$$\|\mathcal{R}\|_\infty = \sup_{u\in[0,2\pi]} \left|\mu\, H[y_u] - y - y\, H[y_u] - H[y\, y_u] + \beta\, \partial_u(y_u/D) - \beta\, H[\partial_u G]\right|,$$

where $D = \sqrt{y_u^2 + (1 + H[y_u])^2}$ and $G = (1 + H[y_u])/D$. Both $\beta = 0$ (M5) and $\beta \neq 0$ (M6) branches mix Path A (exact `TrigPoly` algebra for $H[y_u]$ and other trig-poly arguments) with Path B (certified DFT Hilbert from M4) where the argument is not a trig polynomial. Specifically: $H[y \cdot y_u]$ is routed through Path B even in M5, and $H[\partial_u G]$ joins it in M6. Both Path B calls use the new `fft_hilbert_series` interface, supplying `Gpp_bound` and `Gppp_bound` derivative-margin bounds extracted from extended-order `TaylorSeries` chains. All thirteen milestone tests (T5.1–T5.6, T6.1–T6.7) pass.

Contents

1 The Babenko Residual

The Babenko water-wave equation in conformal variables u reads

$$\mu\, H[y_u] = y + y\, H[y_u] + H[y\, y_u] - \beta\, \partial_u\Big(\frac{y_u}{D}\Big) + \beta\, H[\partial_u G]\,, \tag{1}$$

where $y : \mathbb{T} \to \mathbb{R}$ is the approximate solution, H is the periodic Hilbert transform, $\mu > 0$ is the speed parameter, $\beta \geq 0$ is the (Bond-number-like) surface-tension parameter, and

$$D(u) = \sqrt{y_u(u)^2 + (1 + H[y_u](u))^2}, \qquad G(u) = \frac{1 + H[y_u](u)}{D(u)}.$$

Given a finite cosine series

$$y(u) = \frac{a_0}{2} + \sum_{k=1}^{N} a_k \cos(ku),$$

the *residual* is

$$\mathcal{R}(u) = \mu\, H[y_u](u) - y(u) - y(u)\, H[y_u](u) - H[y\, y_u](u) + \beta\, \partial_u\Big(\frac{y_u}{D}\Big)(u) - \beta\, H[\partial_u G](u). \tag{2}$$

2 Computation Strategy

2.1 Two paths

The residual separates into terms handled by two certified paths:

Path A (exact trig-poly arithmetic). $H[y_u]$ is the only Hilbert application whose argument is a trig polynomial *as a stored object*. It is computed exactly using the M1 `TrigPoly` algebra and converted to certified `TaylorSeries` via `trigpoly_to_taylor`.

Path B (DFT Hilbert, certified aliasing). $H[y{\cdot}y_u]$ and $H[\partial_u G]$ are both routed through Path B. Although $y \cdot y_u$ is mathematically a trig polynomial of degree $\leq 2N$, in the current pipeline it is held as a `TaylorSeries` `ts_y_yu` (product of two `TaylorSeries` objects); applying Path B to it via `fft_hilbert_series` is uniform with the $H[\partial_u G]$ case, which is genuinely non-polynomial. Both Path B calls use the rigorous aliasing bound $\varepsilon_{\text{alias}} = 4C_p/((p{-}1)(N/2-1)^{p-1})$ per ball and the Fourier-tail `sup_bounds` formula $\|\hat{H}[G^{(m)}]\|_\infty \leq \|G^{(m+2)}\|_\infty \cdot \pi^2/3$ for the output series (see M4 report).

2.2 Taylor series representation

All terms are assembled as `TaylorSeries` objects. Recall the key primitives from M2:

- `trigpoly_to_taylor(f, N_grid, M_order)` builds the matrix $[f^{(m)}(u_j)]_{j,m}$ and the full `sup_bounds` vector $\texttt{sup_bounds}[m+1] \geq \|f^{(m)}\|_\infty$ for $m = 0, \ldots, M{+}1$.
- `TaylorSeries` multiplication uses the Leibniz formula at grid points and exact-Leibniz on the `sup_bounds` vectors: $\texttt{sup_bounds}_{fg}[m{+}1] = \sum_{k=0}^{m} \binom{m}{k} \texttt{sup_bounds}_f[k{+}1]\, \texttt{sup_bounds}_g[m{-}k{+}1]$.
- `sup_bound(ts)` returns the certified $\sup|f|$ via the Taylor remainder bound $\texttt{sup_bounds}[\text{end}] \cdot h^{M+1}/(M+1)!$.

2.3 M5 assembly ($\beta = 0$)

1. Build `TrigPoly` objects for y, y_u, $H[y_u]$ (Path A, exact).
2. Convert to `TaylorSeries` at order M: `ts_y`, `ts_yu`, `ts_Hyu`.
3. Multiply: `ts_y_Hyu` = `ts_y`×`ts_Hyu`; `ts_y_yu` = `ts_y`×`ts_yu`.
4. Build the extended-order pair `ts_y_g`, `ts_yu_g` at order $M_h = M + 2$ and multiply to get `ts_y_yu_h`; extract Gpp_bound = `ts_y_yu_h.sup_bounds`$[M{+}3]$ and Gppp_bound = `ts_y_yu_h.sup_bounds`[en These supply the two derivative-margin bounds that `fft_hilbert_series` requires.
5. Path B: `ts_Hyyu` = H[`ts_y_yu`] via `fft_hilbert_series`(`ts_y_yu`, N; `Gpp_bound`, `Gppp_bound`, ...).
6. Assemble: `ts_R` = μ `ts_Hyu`−`ts_y`−`ts_y_Hyu`−`ts_Hyyu`.
7. Return `sup_bound(ts_R)`.

2.4 M6 assembly ($\beta \neq 0$)

Steps 1–7 are identical to M5. Additional steps:

8. Certify D positivity at order $M_1 = M + 1$: $(D_{\text{lower}}, \texttt{ts_D}) = \texttt{dmin_bound(y, N, M_1)}$. If $D_{\text{lower}} \leq 0$, the computation aborts.

9. Invert D: $\texttt{ts_invD} = 1/D$ via `inv_series`.

10. Compute $\partial_u(y_u/D)$: build `ts_yu_M1` at order M_1, multiply `ts_yu_M1`×`ts_invD`, then `deriv_series` to get `ts_C1` at order M.

11. Compute $G = B/D$ where $B = 1 + H[y_u]$: construct `ts_B` at order M_1 from the iterated $H[y_u]$ derivatives, then $\texttt{ts_G} = \texttt{ts_B} \times \texttt{ts_invD}$. Differentiate: $\texttt{ts_Gu} = \partial_u G$ via `deriv_series`, landing at order M.

12. Build the M3-extended chain at order $M_3 = M + 3$: $(D_{\text{lower},3}, \texttt{ts_D3}) = \texttt{dmin_bound(y, N, M_3)}$; $\texttt{ts_invD3} = \texttt{inv_series(ts_D3, } D_{\text{lower},3}\texttt{)}$; `ts_B3` at order M_3 from iterated $H[y_u]$; `ts_G3` = `ts_B3`×`ts_invD3`. Extract Gpp_bound = `ts_G3.sup_bounds`$[M{+}4]$ (bounds $G^{(M+3)} = (G_u)^{(M+2)}$) and Gppp_bound = `ts_G3.sup_bounds`$[M{+}5]$ (bounds $G^{(M+4)} = (G_u)^{(M+3)}$).

13. Certified Hilbert: $\texttt{ts_HGu} = H[\partial_u G]$ via `fft_hilbert_series`(`ts_Gu`, N; `Gpp_bound`, `Gppp_bound`, ...).

14. Assemble capillary: $\texttt{ts_R} \mathrel{+}= \beta\,\texttt{ts_C1} - \beta\,\texttt{ts_HGu}$.

15. Return `sup_bound(ts_R)`.

The extra Taylor orders ($M_1 = M + 1$ for intermediate computations of $\partial_u(y_u/D)$ and $\partial_u G$; $M_3 = M + 3$ for extracting the derivative-margin bounds for the Path B `fft_hilbert_series` call) ensure that after one application of `deriv_series` the capillary terms land at the same order M as the base residual, and that the Fourier-tail bound on $\|\hat{H}[G^{(M)}]\|_\infty$ has two derivative-margin bounds available.

3 Implementation Notes

3.1 D positivity sharing

Computing y_u/D and $G = (1 + H[y_u])/D$ both require D^{-1}. Rather than calling `compute_G_taylor` (which re-runs `dmin_bound` internally), the M6 branch calls `dmin_bound` once and shares `ts_invD` between y_u/D and $G = B \cdot D^{-1}$. This halves the cost of the most expensive step.

3.2 sup_bound robustness

An issue was discovered and fixed in `sup_bound`: the original code used `Arblib.is_positive(s - best)` to track the running maximum, but when the `TaylorSeries` derivs are Arb balls that straddle zero (a common occurrence at small N_{fft} where the DFT aliasing radius is large), `abs(arb_ball)` returns a ball with lower bound 0, making `is_positive` fail. The fix: compare Float64 upper bounds of s and `best`, which is valid because `ubound`(s) is a certified upper bound on the Taylor expansion for that subinterval, and taking the max of upper bounds gives a valid certified upper bound on the supremum.

3.3 Tightness of bounds

The certified/Float64 ratio measures over-estimation. For $\beta = 0$ (M5, random seeds) the ratio is 1.015–1.040, showing the bound is essentially tight. For $\beta \neq 0$ (M6, random seeds) the ratio varies considerably with the input: for the smooth linear cases it is $\approx$ 1.1–3.0, but for high-frequency random inputs it can blow up to 10^3–10^7. This is the genuine cost of correctness for `inv_series` and `sqrt_series`: the previous formulas silently ignored intermediate-derivative bounds (audit C1–C2) and were tighter than the truth would allow; the new Faà di Bruno composition correctly tracks derivative growth through the Bell-polynomial expansion, and high-frequency inputs incur genuinely large bounds on $\|(1/f)^{(M+1)}\|_\infty$ and $\|(\sqrt{f})^{(M+1)}\|_\infty$. For the specific smooth candidates used in the existence proof, the bound remains tight.

4 Milestone Tests

4.1 M5 Tests ($\beta = 0$)

T5.1 — Small-amplitude Stokes wave

Input: $y = -\varepsilon^2/2 + \varepsilon\cos u + \varepsilon^2\cos 2u$, $\mu = 1 + \varepsilon^2$, $\varepsilon = 0.15$, $\beta = 0$.

$$\text{Certified bound} = 0.01750 \leq 0.05. \quad \checkmark$$

T5.2 — Containment

Float64 estimate via direct DFT on $N = 512$ grid: 1.744×10^{-2}. Certified bound $1.750 \times 10^{-2} \geq 1.744 \times 10^{-2}$. ✓

T5.3 — Convergence in ε

ε	Certified bound
0.01	2.85×10^{-5}
0.05	1.06×10^{-3}
0.10	6.02×10^{-3}
0.15	1.75×10^{-2}

Bounds increase monotonically with ε. ✓

T5.4 — Convergence in Taylor order

M_{order}	Certified bound
2	1.802×10^{-2}
4	1.750×10^{-2}
6	1.7499×10^{-2}

Bound decreases then plateaus: the residual itself is the floor. ✓

T5.5 — $\beta = 0$ branch

$\beta = 0$ returns a positive, finite bound. ✓

T5.6 — Random containment (10 seeds)

All 10 certified bounds contain the Float64 estimate. Ratio range: $[1.015, 1.040]$. ✓

4.2 M6 Tests ($\beta \neq 0$)

T6.1 — Linear gravity-capillary mode

Input: $y = 0.05\cos u$, $\mu = 1 + \beta = 1.05$, $\beta = 0.05$, $N_{\text{grid}} = 256$.

$$\text{Certified bound} = 3.696 \times 10^{-3} \leq 0.02. \quad ✓$$

$D_{\text{lower}} = 0.950 > 0$, so the D-positivity certificate holds. The bound is materially tighter than the previous M6 result of 5.58×10^{-3} at the same N: this is the direct consequence of the Fourier-tail $\|\hat{H}[G^{(M)}]\|_\infty \leq \|G^{(M+2)}\|_\infty \cdot \pi^2/3$ formula replacing the (unsound) Lebesgue-constant estimate (audit C1). For smooth G the $\pi^2/3 \approx 3.29$ factor is much smaller than the old $2\log N + 2 \approx 13.1$ at $N = 256$.

T6.2 — $\beta = 0$ recovery

Passing $\beta = 0$ to the M6 code returns the identical bound as M5. (The M5 and M6 branches coincide at $\beta = 0$.) ✓

T6.3 — D positivity certified

$$D_{\text{lower}} = 0.950 > 0.01. \quad ✓$$

T6.4 — Aliasing subdominant at large N_{fft}

N_{fft}	Certified bound
256	3.696×10^{-3}
2048	3.691×10^{-3}

At $N = 256$ the certified bound is already within 0.15% of the $N = 2048$ value, confirming the aliasing contribution is essentially saturated. (With the previous Lebesgue-constant formula the gap was $\approx 50\%$.) ✓

T6.5 — Convergence in N_{fft}

N_{fft}	Certified bound
128	3.714×10^{-3}
256	3.696×10^{-3}
512	3.692×10^{-3}
1024	3.691×10^{-3}

Bounds decrease monotonically as N increases, converging to the residual floor as the Path B aliasing radius shrinks. ✓

T6.6 — Random containment, $\beta \neq 0$ (10 seeds)

All 10 certified bounds contain the Float64 estimate. Ratio range: $[1.12, 7.7 \times 10^6]$, with the large values arising for high-frequency random inputs (large N modes); this is the genuine cost of correctness in `inv_series`/`sqrt_series` no longer ignoring derivative bounds. For the smooth candidates relevant to the existence proof the ratios are ≤ 5. The test additionally asserts `all(isfinite, ratios)` and `maximum(ratios) < 1e8` so silent regressions to vacuous / infinite bounds are caught. ✓

T6.7 — $\beta \neq 0$ with $M_{\text{order}} = 1$ (smallest legal)

Audit gap closed. All M6 tests above use $M_{\text{order}} = 4$; T6.7 exercises the edge case $M_{\text{order}} = 1$ where the extended chains collapse to $M_1 = 2$ and $M_3 = 4$. Input: same Stokes-like $y = 0.05 \cos u$, $\mu = 1.05$, $\beta = 0.05$, $N_{\text{grid}} = 256$.

$$\text{Certified bound} = 4.475 \times 10^{-3} \leq 0.05. \quad ✓$$

The bound is larger than T6.1's 3.696×10^{-3} at $M_{\text{order}} = 4$ (fewer Taylor terms means looser remainder), as expected. ✓

5 Summary and Next Steps

`ResidualBound.jl` now supports both the pure-gravity ($\beta = 0$) and gravity-capillary ($\beta \neq 0$) Babenko residual. The key numbers:

Milestone	Primary test case	Certified bound
M5 ($\beta = 0$)	Stokes wave $\varepsilon = 0.15$	1.750×10^{-2}
M6 ($\beta = 0.05$)	Grav-cap $\varepsilon = 0.05$, $N_{\text{grid}} = 256$	3.696×10^{-3}

For the smooth candidates relevant to the existence proof, the certified/Float64 ratio is $\leq 5\times$ throughout. High-frequency random inputs in T6.6 can produce much looser ratios; these reflect the genuine growth of $\|(1/D)^{(M+1)}\|_\infty$ for non-smooth inputs and are not a defect of the bound.

The next milestone is M7 (`Certificate.jl`): writing a Lean-parseable certificate file from the Julia output, enabling the Lean proof of the bound.